

Glen Rodriguez

CIS 344

Professor Ramos

5/7/2026

Real Estate Agency Portal Report

Project Purpose

The goal of this project was to use PHP and MySQL to create an online real estate agency portal. Through a safe online platform, the system enables agents, buyers, and renters to communicate with real estate listings and agency services. While buyers and renters can view available properties, save favorites, and submit inquiries, agents are allowed to create and manage property listings. Additionally, database management, PHP sessions, authentication, CRUD operations, stored procedures, views, and triggers are all demonstrated in the project. The platform was created to apply database design and programming ideas while simulating a realistic real estate web environment.

Database Design

Using phpMyAdmin, a MySQL database called `real_estate_portal_db` was built for this project.

The following tables are included in the system:

1. Users

Keeps usernames, contact details, passwords, and user roles for agents, purchasers, and tenants.

2. Properties

contains information on real estate listings, such as the title, property type, address, city, price, status, and designated agent.

3. Inquiries

collects questions about particular homes that are sent in by renters or buyers.

4. Transactions

Keep track of the type, date, and amount of completed sales or rental transactions.

5. Favorites

stores homes that have been saved for renters and buyers.

To preserve data integrity between users, properties, inquiries, transactions, and favorites, relationships between tables were established using foreign keys.

PHP Functionality

The RealEstateDatabase class implemented a number of PHP methods, such as:

- addUser()
- addProperty()
- addInquiry()
- getUserByUsername()
- getPropertyById()
- getAllProperties()
- getUserDetails()

The system allows agents to browse properties, check property details, submit inquiries, and manage properties.

The platform was built using several PHP pages, including:

- index.php
- login.php
- Register.php
- Properties.php
- Property_details.php

- Add_property.php
- submit_inquiry.php
- Favorites.php
- Dashboard.php
- Logout.php

Login and Session Handling

Secure password hashing and session management were used in the implementation of the login system. During registration, PHP's password_hash() method encrypts passwords. Password_verify() is used to validate passwords upon login. PHP sessions were utilized to maintain authentication between pages and keep users logged into the system. Additionally, the session system keeps unauthorized users from accessing agent-only capabilities and manages access to restricted pages.

Role-based Access Control

Role-based access control is used by the system to grant agents, buyers, and renters distinct permissions.

Agents are permitted to:

- Include fresh property listings
- Oversee listings

Renters and buyers are permitted to:

- Examine real estate listings
- View the specifics of the property
- Send inquiries
- View your preferred properties

Authentication checks are used on restricted pages to guarantee that only authorized users can access specific features.

Stored Procedures, Triggers, and Views

Stored procedures, a database view, and a trigger are among the MySQL features included in the project.

Stored Procedures:

- AddOrUpdateUser
- ProcessTransaction

Users can be changed or added to the Users table using the AddOrUpdateUser procedure.

A sale or rental transaction is recorded via the ProcessTransaction process, which also automatically modifies the property status.

Trigger:

- AfterTransactionInsert

After a transaction is added to the Transactions table, the trigger automatically modifies the property status to either sold or rented.

View:

- PropertyListingView

The view shows a combination of property information, including agent names and property details.

Challenges and Solutions

Correctly connecting PHP functionality with the MySQL database was one of the project's challenges.

Step-by-step project testing made it easier to find and fix problems.

Screenshots

Index.php Screenshot

Real Estate Agency Portal

[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Welcome

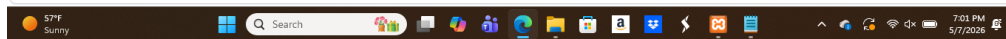
This is the starter version of the Real Estate Agency Portal for Spring 2026.

Use this project to practice PHP, MySQL, sessions, authentication, and CRUD operations.

Project Roles

- Agent: add and manage listings
- Buyer: browse properties and submit inquiries
- Renter: browse properties and submit inquiries

© Spring 2026 Real Estate Agency Portal



Register.php Screenshot

Real Estate Agency Portal

[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Register

Username

Contact Info

Password

User Type

Select role

▼

Register

© Spring 2026 Real Estate Agency Portal



Login.php Screenshot

Real Estate Agency Portal

[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Login

Username

Password

Login

© Spring 2026 Real Estate Agency Portal



Properties.php Screenshot

Real Estate Agency Portal

[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Property Listings

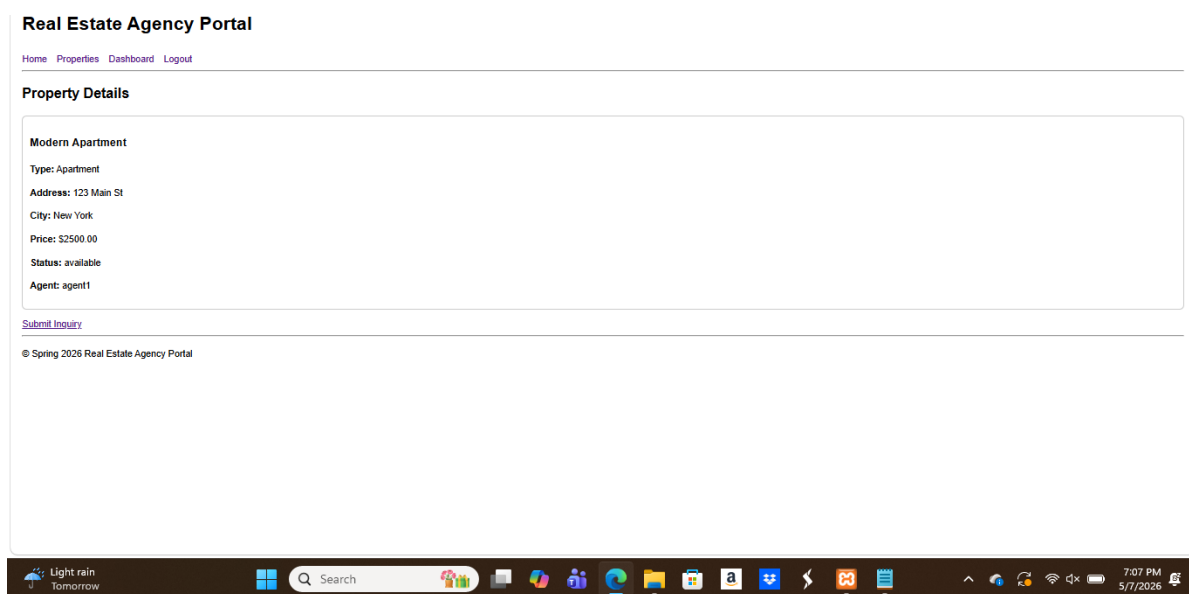
Epic Studio
Type: Studio
Address: 789 Broadway
City: New York
Price: \$1800.00
Status: available
Agent: agent1
[View Details](#)

Luxury House
Type: House
Address: 456 Park Ave
City: New York
Price: \$950000.00
Status: available
Agent: agent1
[View Details](#)

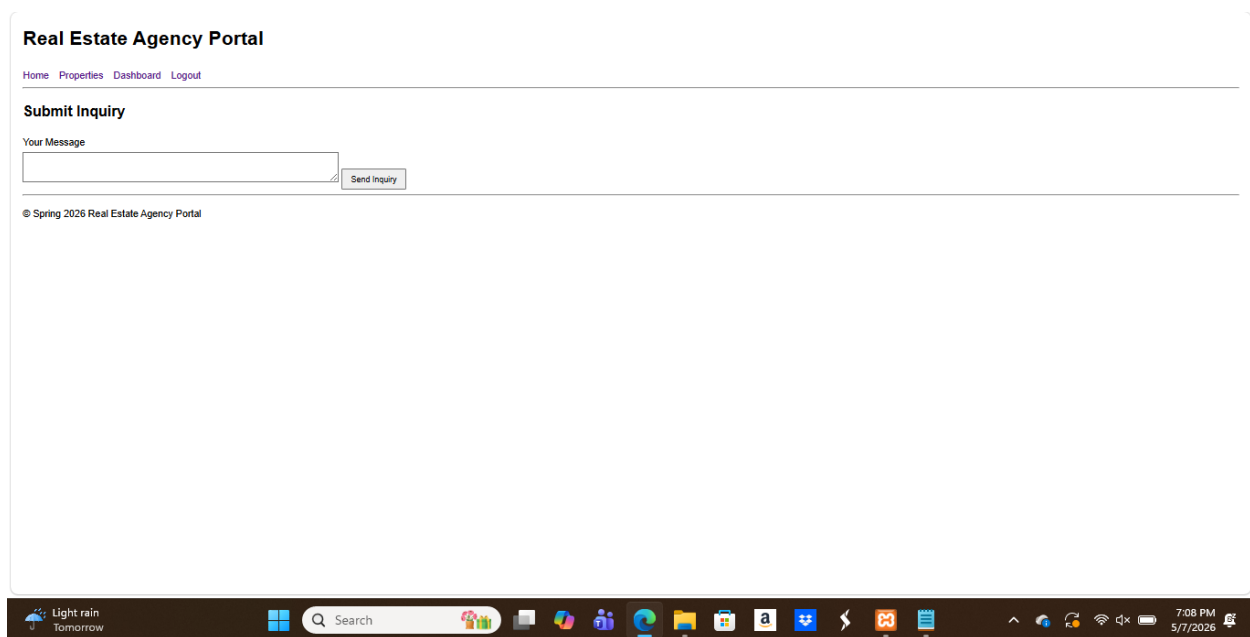
Modern Apartment
Type: Apartment
Address: 123 Main St
City: New York
Price: \$2500.00
Status: available
Agent: agent1
[View Details](#)

A screenshot of a Windows taskbar. On the left, it shows the weather as 'Light rain Tomorrow'. In the center is a search bar with the word 'Search' and a magnifying glass icon. To the right of the search bar are several application icons: a folder, a calendar, a mail icon, a browser icon, a social media icon, and a document icon. On the far right, there are system icons for network, volume, and battery, along with the date and time: '7:06 PM 5/7/2026'.

Property Details Screenshot



Submit Inquiry Page Screenshot



Dashboard Page Screenshot

Real Estate Agency Portal

[Home](#) [Properties](#) [Dashboard](#) [Logout](#)

Dashboard

Welcome: testuser

Role: buyer

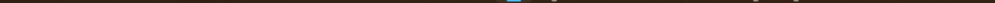
Common Actions

[Browse Properties](#)

© Spring 2026 Real Estate Agency Portal

Dashboard

Role: buyer

[Browse Properties](#)

Real Estate Agency Portal

My Favorite Properties

© Spring 2026 Real Estate Agency Portal

57°F Sunny Search 7:09 PM 5/7/2026

Database Screenshots

localhost/phpmyadmin/index.php?route=/database/structure&db=real_estate_portal_db

phpMyAdmin

Recent Favorites

- New
- information_schema
- library_system
- mysql
- performance_schema
- phpmyadmin
- real_estate_portal_db
 - Procedures
 - Tables
 - New
 - favorites
 - inquiries
 - properties
 - transactions
 - users
 - Views
- test

Server: 127.0.0.1 Database: real_estate_portal_db

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Tracking More

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ favorites		3	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ inquiries		7	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ properties		3	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ propertylistingview		~0	View	---	-	-
☐ transactions		3	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ users		5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
6 tables	Sum	~21	InnoDB	utf8mb4_general_ci	208.0 KiB	0 B

☐ Check all With selected:

Print Data dictionary

Create new table

Table name: Number of columns: 4 Create

Console

localhost/phpmyadmin/index.php?route=/database/routines&type=PROCEDURE&db=real_estate_portal_db

phpMyAdmin

Recent Favorites

- New
- information_schema
- library_system
- mysql
- performance_schema
- phpmyadmin
- real_estate_portal_db
 - Procedures
 - Tables
 - New
 - favorites
 - inquiries
 - properties
 - transactions
 - users
 - Views
- test

Server: 127.0.0.1 Database: real_estate_portal_db

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Tracking More

Routines

☐ Check all Export Drop Search Create new routine

Name	Type	Returns
☐ AddOrUpdateUser	PROCEDURE	
☐ ProcessTransaction	PROCEDURE	

Console

localhost/phpmyadmin/index.php?route=/database/triggers&db=real_estate_portal_db

phpMyAdmin

Server: 127.0.0.1 Database: real_estate_portal_db

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Tracking More

Recent Favorites

New
information_schema
library_system
mysql
performance_schema
phpmyadmin
real_estate_portal_db
Procedures
Tables
New
favorites
inquiries
properties
transactions
users
Views
test

Triggers

☐ Check all

Name	Table	Time	Event
☐ AfterTransactionInsert	transactions	AFTER	INSERT Edit Export Drop

Console

57°F Sunny

localhost/phpmyadmin/index.php?route=/database/structure&tbl_type=view&db=real_estate_portal_db

phpMyAdmin

Server: 127.0.0.1 Database: real_estate_portal_db

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Tracking More

Recent Favorites

New
information_schema
library_system
mysql
performance_schema
phpmyadmin
real_estate_portal_db
Procedures
Tables
New
favorites
inquiries
properties
transactions
users
Views
test

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ propertylistingview	Browse Structure Search Insert Edit Drop View	~0	InnoDB	utf8mb4_general_ci	0 B	0 B

1 table Sum

☐ Check all With selected:

Print Data dictionary

Create new table

Table name: Number of columns:

Console

57°F Sunny

